FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

FULL TEST – I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 234**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

   Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (13 to 18) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

2. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A
Straight Objective Type

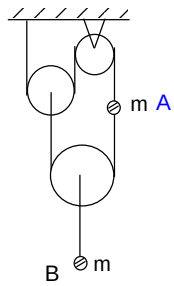
This section contains **08 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. A ball is dropped from rest at height 4h. After it has fallen a distance d, a second ball is dropped from rest at height h. What should d be (in terms of h) so that the balls hit the ground at the same time?

(A) $d = \frac{3H}{2}$ (B) $d = \frac{H}{3}$

(C) d = H (D) d = 3H

2. What is the acceleration of the upper block of the system as shown in the figure? Assume pullies and strings are ideal. Given $m_A \simeq 0$



(A) 4g (B) 2g

(C) g (D) $\frac{g}{2}$

3. The pitch of a screw gauge is 0.5 mm and there are 50 divisions on its circular scale and one main scale division = 0.5 mm. Before starting the measurement it is found that when jaws of the screw gauge are brought in contact, the zero of the circular scale lies 4 division above the reference line. When a metallic wire is placed between the jaws, five main scale divisions are clearly visible and 18th division on the circular scale coincides with the reference line. The diameter of the wire is

(A) 2.68 mm (B) 2.72 mm

(C) 2.64 mm (D) 2.62 mm

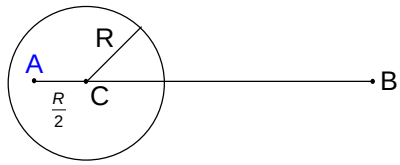
Space for rough work

4. A point like particle of mass 'm' (very small) is projected in the vertically upward direction where already exist, a uniform horizontal electric field $\vec{E}$. The field strength is such that qE= $\frac{3}{4}$ mg, where q is the charge on the particle. After what time the radius of curvature of the charged particle will be minimum.

(A) $\frac{V_0}{g}$ (B) $\frac{8V_0}{25g}$

(C) $\frac{16V_0}{25g}$ (D) $\frac{25V_0}{16g}$

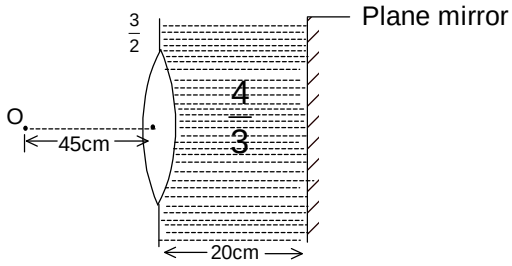
5. Consider a thin conducting shell of radius R carrying total charge Q .Two Point charges Q and 2Q are placed on points A and B, which are at a distances of $\frac{R}{2}$ and 2R from the center of the cell respectively as shown in the figure. If the cell is earthed how much charge will flow to the earth?



(A) 3Q (B) Q

(C) 2Q (D) $\frac{Q}{2}$

6. Figure shows an object 'o' placed in front of a biconvex lens $\left(\mu = \frac{3}{2}\right)$ behind which there is water $\left(\mu = \frac{4}{3}\right)$ and a plane mirror at a distance of 20m. A point object 'O' is placed at a distance of 45 cm from the lens as shown in the figure .The position of the image of the object coincides with the object itself, then radius of curvature of the lens is.



(A) 20cm
(B) 45cm
(C) 60cm
(D) 30cm

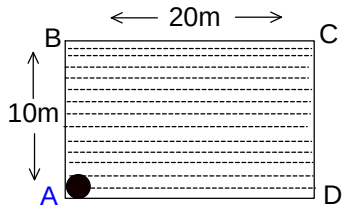
Space for rough work

7. A satellite of mass m is orbiting the earth in a circular orbit of radius r. It starts losing its mechanical energy due to small air resistance at the rate of k joule/sec .The time taken by the satellite to hit the surface of the earth is (M and R are the mass and radius of the earth)

(A) $\frac{GMm}{K}\left(\frac{1}{R}-\frac{1}{r}\right)$ (B) $\frac{GMm}{2K}\left(\frac{1}{R}-\frac{1}{r}\right)$

(C) $\frac{2GMm}{K}\left(\frac{1}{R}-\frac{1}{r}\right)$ (D) $\frac{GMm}{K}\left(\frac{1}{R}+\frac{1}{r}\right)$

8. There is a container ABCD which is accelerating towards right, at the rate of $\frac{g}{2} m/s^2$. Container is filled completely by a liquid of relative density 3. A small ball of relative density 1 is released from rest with respect to the container at point A of the container. At what distance from the end point B, the ball will hit to the top surface of the container.

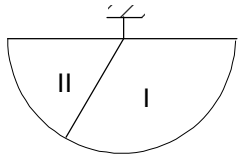


(A) 2m (B) 5m
(C) 15m (D) 10m

Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

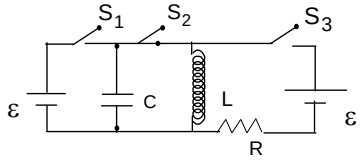
9. A pendulum swings back & forth between two horizontal positions as shown in the figure, under uniform gravity, then



(A) There are more than one points in the part I where acceleration is vertical.
(B) There is at least one point in part I where acceleration is horizontal.
(C) During the motion in both parts tension is maximum only once.
(D) Acceleration of the bob is never directed towards the point of suspension O.

Space for rough work

10. In the arrangement as shown in the figure, S_1 and S_3 are already closed for a long time. Now at t = 0 the S_2 is closed and S_1, S_3 are opened. [ε = 1 volt, L = 1 henry, R = 1 Ω & C = 1 Faraday]



(A) After S_2 is closed the maximum current through the indictor is 2A.

(B) After S_2 is closed the maximum current through the indictor is $\sqrt{2}A$

(C) Maximum charge on the capacitor is $2+\sqrt{2}$ coulomb

(D) Maximum charge on the capacitor is $\sqrt{2}$ coulomb

11. A particle of mass m = 3 kg is moving on a straight line in a conservative force field in which the potential energy associated with the particle is given by $U(y) = 2y^2 - 30y + 450$. At t = 0, particle is at y = 8 m and moving towards positive y-axis with 1 m/s.

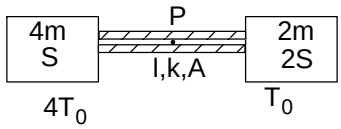
(A) During the course of its motion the maximum speed of the particle will be $\sqrt{\frac{4}{3}}$ m/s.

(B) Particle will never cross the origin.

(C) Particle will come to its equilibrium position for the first time at $\frac{5\pi}{4\sqrt{3}}$ *second*

(D) The amplitude of oscillation for the particle is 1m.

12. Figure shows two reservoirs containing two liquids of masses 4 m and 2 m and their specific heat capacities are S and 2S respectively. Their initial temperatures are $4T_0$ and T_0 respectively. The containers are joined by a conducting rod of thermal conductivity K, length l and cross section area A specific heat capacity of the rod is negligible.



(A) The temperature of mid-point 'P' of the rod is always constant.

(B) Heat extracted from container 1 = Heat supplied to container 2

(C) After time $t_0 = \frac{2msl}{ka}\ln 2$ the temperature difference between containers becomes half of the initial temperature difference.

(D) After time $t_0 = \frac{msl}{ka}\ln 2$ the temperature difference between containers becomes half of the initial temperature difference.

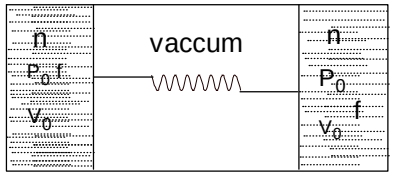
Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

Paragraph for Questions 13 & 14

In the arrangement shown in the fig the cylinder is insulating one. Both sides same diatomic gas is trapped by two insulting massless pistons with the help of an ideal spring. The natural length of the spring is equal to the length of the cylinder. Initial state of the gases are as shown in the figure.



13. What is the value of energy stored in the spring?
(A) P_0V_0 (B) $2P_0V_0$
(C) $4P_0V_0$ (D) $\frac{P_0V_0}{2}$

14. Now the gases are heated slowly, such that their temperature becomes three times to their initial temperature. The total heat given to the system is
(A) $12P_0V_0$ (B) $8P_0V_0$
(C) $5P_0V_0$ (D) $6P_0V_0$

Paragraph for Questions 15 & 16

Initially the nucleus of radium-226 is at rest. It decays due to which and α particle and the nucleus of radon are created. The released energy during the decay is 4.87 Mev, which appears as the kinetic energy of the two resulted particles. $[m_\alpha = 4.002amu, m_{Rn} = 222.017amu,]$

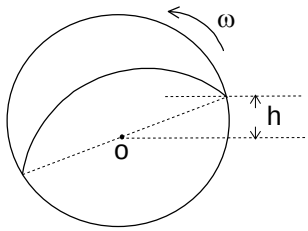
15. Kinetic energies of α particle & radon nucleus are respectively
(A) 0.09 Mev, 4.08 Mev (B) 4.78 Mev , 0.09 Mev
(C) 4.08 Mev , 0.09 Mev (D) 3.68 Mev , 0.08 Mev

16. What is the linear momentum of the α – particle?
(A) 2.01×10^{-19}kgms^{-1} (B) 1.01×10^{-19}kgms^{-1}
(C) 4.01×10^{-19}kgms^{-1} (D) 8.01×10^{-19}kgms^{-1}

Space for rough work

Paragraph for Questions 17 & 18

There is a hollow cylinder of radius R and it is rotating with constant angular speed ω. There is a point mass which rotates along with the cylinder and gets carried upward. Friction is enough such that the point mass does not slip with respect to the cylinder as long as the normal force is becomes zero. If the path of the particle after it lost contact with the cylinder is as shown in the figure then,



17. In Order to happen this where most the particle lose contact with the cylinder?

(A) $h = \frac{R}{2}$ (B) $h = \frac{R}{\sqrt{3}}$

(C) $h = \frac{R}{\sqrt{2}}$ (D) $h = \frac{R}{\sqrt{\pi}}$

18. What is the value of ω (in terms of g and R)

(A) $\sqrt{\frac{g}{\sqrt{3}R}}$ (B) $\sqrt{\frac{g}{2\sqrt{\pi}R}}$

(C) $\sqrt{\frac{g}{2R}}$ (D) $\sqrt{\frac{g}{\sqrt{2}R}}$

SECTION – C
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

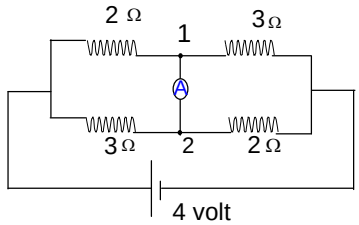
19. A square of elastic sheet of dimension (a × a) has a mass 125 gm. A force of 2.5 N is applied to each of the four edges. What is the velocity of waves on the sheet? a = 0.75 m.

20. In a Young's double slit experiment, the angular separation of the interference fringes on a distant screen is $0.04A^0$. The angular separation if the entire apparatus is immersed in a liquid of R.I. $\frac{4}{3}$ is $0.0n\ A^0$, then what is the value of n?

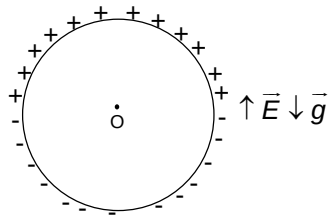
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21. In the circuit shown, the reading of the ammeter is $\frac{p}{q}$ amp.

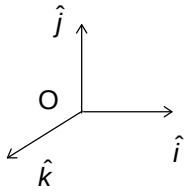
What is the value of p + q?

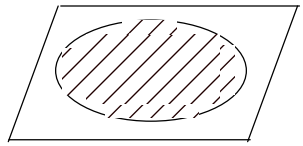


22. In the figure shown there is non- conducting disc of mass M = 2kg and radius R = 4 m. On its upper & lower part of circumference +Q and –Q charge are uniformly attached such that liner charge density is $\frac{Q}{\pi R}$. The disc can freely rotate about an horizontal axis passing through O. There is a uniform electric field $\bar{E}$ in the vertical direction such that QE= mg. If the disc is rotated by a small angle it performs S.H.M. its time period is given by $T \simeq \sqrt{\frac{p\pi}{n}}$ then, P + n is. (Take $\pi^2 = g$)



23. There is a non-conducting disc of mass 4kg and radius 1m placed on a rough non conducting surface. A conducting ring of same mass is tightly fixed around the disc. There exists a magnetic field $\vec{B} = 4\hat{i} + 4t^2\hat{j}$. The resistance of the ring is 8Ω. Find the time after which the system will start toppling.





Space for rough work

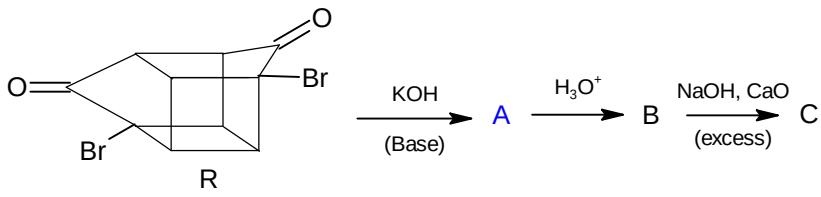
Chemistry

PART – II

SECTION – A
Straight Objective Type

This section contains **08 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

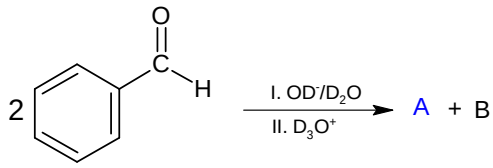
24. Which of the following is/are incorrect statement (s):
(I) $CHCl_3$ is less acidic than CHF_3
(II) Dipole moment of chlorobenzene is more than chlorocyclohexane
(III) Dipole moment of fluoromethane is more than chloromethane
(IV) $NH_2NH_2.HCl$ show positive test for nitrogen in lassaigne's test
(V) Dipole moment of difluoromethane is more than dichloromethane.
(A) I Only (B) II Only
(C) I, II, III and IV (D) I, II, III, IV and V

25. 

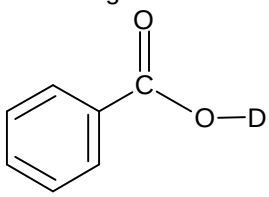
Which of the following is correct statement?
(A) Monobromination of C gives two product. (B) Degree of unsaturation of C is five.
(C) Degree of unsaturation of B is 8 (D) None of these.

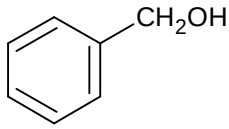
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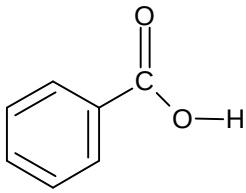


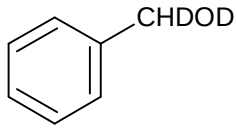
Which of the following statement is correct?

(A) 

A is 

and B is

(B) 

A is 

and B is

(C) Order of reaction of I is 4 at very high concentration of OD^-

(D) Order of reaction of I is 3 at very high concentration of OD^-

27. Which of the following statement is incorrect?

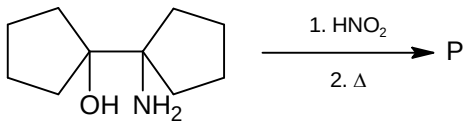
(A) $Cr(CO)_6 < Fe(CO)_5 < Ni(CO)_4$ (Order of increasing CO bond strength.)

(B) $[V(CO)_6]^- < [Cr(CO)_6] < [Mn(CO)_6]^+$ (order of increasing M – C bond strength where M is central metal atom/ion)

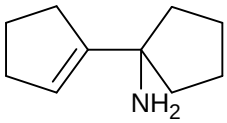
(C) $[V(CO)_6]^- = [Cr(CO)_6] = [Mn(CO)_6]^+$ (Effective atomic no)

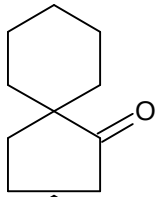
(D) $[V(CO)_6]^- < [Cr(CO)_6] < Mn(CO)_6]^+$ (Order of increasing CO bond strength)

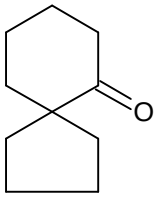
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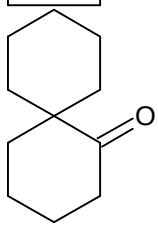
28. 

The P is

(A) 

(B) 

(C) 

(D) 

29. An element undergoes a reaction as shown

$x + 2e^- \rightarrow x^{-2}$

Energy released = 30.87 ev/atom. If the energy released, is used to dissociate 4 g of H_2 molecules equally into H^+ and H^* where H^* is excited state of H atoms where the electron travels in orbit whose circumference equal to four times its de – Broglie's wavelength. Determine the minimum number of moles of x that would be required

Given: IE of H = 13.6 ev/atom, bond energy of H_2 = 4.526ev/molecule

(A) 1 (B) 2

(C) 3 (D) 4

30. How much Ag Br could be dissolved in 1 litre of 0.40 M NH_3? Assume that only complex formed is $[Ag(NH_3)_2]^+$. Given $K_{sp}(AgBr) = 5.0 \times 10^{-13}$; K_f of $[Ag(NH_3)_2]^+ = 1.0 \times 10^8$

(A) 2.82×10^{-3} M (B) 2.22×10^{-4}

(C) 2.22×10^{-3} (D) 2.82×10^{-4}

31. If E_1^0 is standard electrode potential for Fe/Fe^{+2} and E_2^0 is for Fe^{+2}/Fe^{+3} and E_3^0 for Fe/Fe^{+3}. which of the following relation is correct?

(A) $E_3^0 = E_1^0 + E_2^0$ (B) $E_3^0 = \left(E_1^0 + E_2^0\right)/2$

(C) $E_3^0 = E_2^0 + 2E_1^0$ (D) $E_3^0 = \left(E_2^0 + 2E_1^0\right)/3$

Space for rough work

Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

32. A salt MX has NaCl type lattice in which r is the minimum distance between M^{z+} and X^{z-}. Each M^{z+} ion is surrounded by

(A) 6 X^{z-} ions, each at a distance r
(B) 12 M^{z+} ions, each at a distance $\sqrt{2}$ r.
(C) 8 X^{z-} ions, each at a distance $\sqrt{3}$ r.
(D) 12 M^{z+} ions, each at a distance $\sqrt{4}$ r.

33. "$a\ MnO_4^- + b\ H_2O_2 + c\ H^+ \rightarrow d\ Mn^{+2} + e\ O_2 + f\ H_2O$" is a balanced chemical equation (reaction). The values of a, b, c, d, e and f are

(A) 2, 1, 6, 2, 3 and 4 respectively
(B) 2, 3, 6, 2, 4 and 6 respectively
(C) 2, 7, 6, 2, 6 and 10 respectively
(D) None of these

34. $O_{(g)}^- + e^- \rightarrow O_{(g)}^{-2} \quad \Delta H_1$

$O_{(g)} + e^- \rightarrow O_{(g)}^- \quad \Delta H_2$

$O_{2(g)} \rightarrow O_{2(g)}^+ + e^- \quad \Delta H_3$

$O_{(g)} \rightarrow O_{(g)}^+ + e^- \quad \Delta H_4$

$H_{2(g)} \rightarrow H_{2(g)}^+ + e^- \quad \Delta H_5$

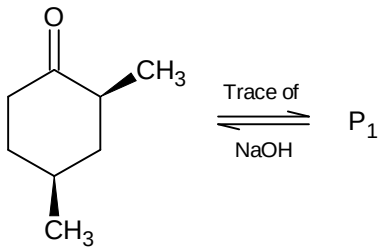
$H_{(g)} \rightarrow H_{(g)}^+ + e^- \quad \Delta H_6$

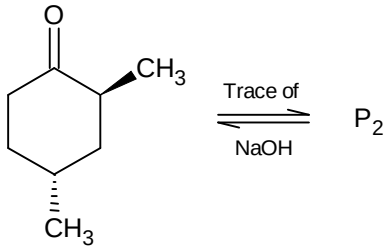
$O_{(g)} + 2e^- \rightarrow O^{-2} \quad \Delta H_7$

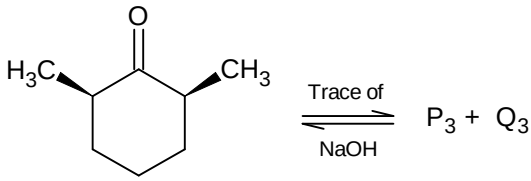
Which of the following option(s) is/are correct?

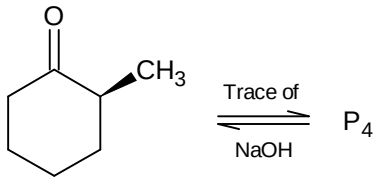
(A) $|\Delta H_1| > |\Delta H_2|$
(B) $|\Delta H_4| > |\Delta H_3|$
(C) $|\Delta H_5| > |\Delta H_6|$
(D) ΔH_7 is +ve

Space for rough work

35. (i) 

(ii) 

(iii) 

(iv) 

All products are in keto form .The correct statement(s) is/are.
(A) P_1 and P_2 are diastereomeric pair.
(B) P_1 and P_2 are identical compound.
(C) In reaction (iv) reactant and product are enantiomeric pair.
(D) P_3 and Q_3 are optically active stereo isomers and both are enantiomeric pair.

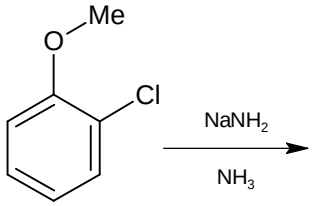
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Comprehension Type

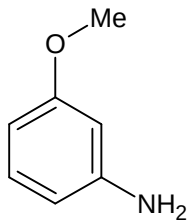
This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

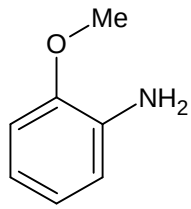
Paragraph for Questions 36 & 37

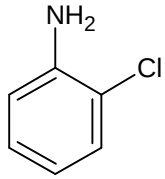
Aryl halides undergo nucleophilic substitution reaction only in extreme condition. Except to certain industrial processes where very severe conditions are feasible, one does not ordinarily prepare phenols (ArOH), ethers (ArOR), amines ($ArNH_2$) or nitriles (ArCN) by nucleophilic attack on aryl halides . The presence of electron withdrawing groups like $–NO_2$, $-CF_3$ at ortho or para position to the halogen atom makes the aryl halides more susceptible to nucleophilic attack. The reaction of unactivated or deactivated aryl halide with strong bases or at high temperature proceed via the benzyne intermediate.

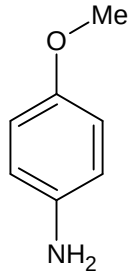
36. 

Product. The predominant product is

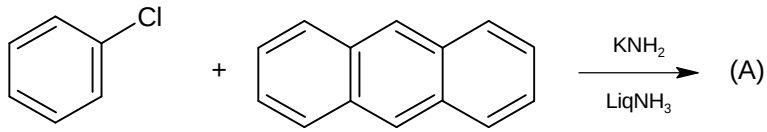
(A) 

(B) 

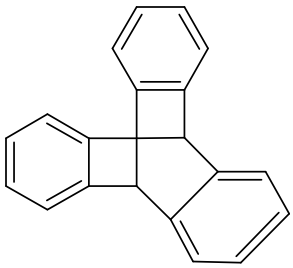
(C) 

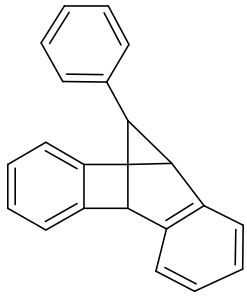
(D) 

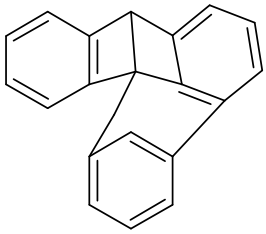
Space for rough work

37. 

Identity A in the above reaction

(A) 

(B) 

(C) 

(D) none of these

Space for rough work

Paragraph for Questions 38 & 39

Ideal gas equation is represented as PV = nRT. Gases present in universe were found ideal in the Boyle's temperature range only and deviated more from ideal gas behavior at high pressure and low temperature. The deviation are explained in term of compressibility factor z. For ideal behavior $Z = \frac{PV}{nRT} = 1$. The main cause to show deviation were due to wrong assumptions made about forces of attractions (which becomes significant at high pressure) and volume V occupied by molecules in PV = nRT is supposed to be volume of gas or the volume of container in which gas is placed by assuming that gaseous molecules do not have appreciable volume. Actually volume of the gas is that volume in which each molecule of gas can move freely. If volume occupied by gaseous molecule is not negligible, then the term V would be replaced by the ideal volume which is available for free motion of each molecule of gas in 1 mole gas.

V_{actual} = volume of container – volume occupied by molecules

= v – b

Where, b represent the excluded volume occupied by molecules present in one mole of gas.

Similarly for n mole gas

V_{actual} = v - nb

38. As the pressure approaching zero ie. at very low pressure. The curves plotted between compressibility factor Z and P for n mole of gases have the following characteristics.
(I) The intercept on y axis leads to a value of unity
(II) The intercept on y axis leads to a value of 'n'.
(III) The curves possess same slope for different gases at same temperature.
(IV) The curves possess different slopes for different gases at same temperature.
(V) The curves possess same slope for a gas at different temperature.
(A) I, IV, V (B) II, III
(C) I, II, III, IV (D) II, III, V

39. The ratio of coefficient of thermal expansion $\alpha = \frac{\left(\frac{\partial V}{\partial T}\right)_P}{V}$ and the isothermal compressibility $\beta = -\frac{\left(\frac{\partial V}{\partial P}\right)_T}{V}$ for an ideal gas is:
(A) $-\frac{P}{T}$ (B) $\frac{P}{T}$
(C) $\frac{T}{P}$ (D) $-\frac{T}{P}$

Space for rough work

Paragraph for Questions 40 & 41

A Colorless solid A liberates a brown gas B on acidification, a colorless alkaline gas (C) on treatment with NaOH and a colorless non-reactive gas D on heating. If heating of the solid is continued, it completely disappears.

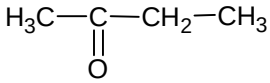
40. The compound A is
(A) NH_4NO_2 (B) PH_4NO_3
(C) NH_4NO_3 (D) none of these

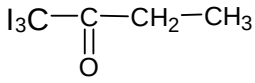
41. Which one of the following is correct?
(A) Bond order of D is 2.5
(B) D is a paramagnetic gas
(C) Smell of gas c is like rotten fish
(D) Ionization energy of gas D is more than N atom

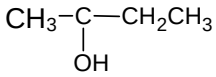
SECTION – C
(One Integer Value Correct Type)

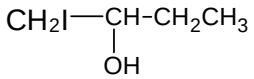
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

42. Find out no of compounds that gives positive iodoform test (either in cold or hot condition)

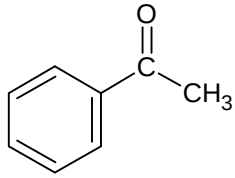
(A) 

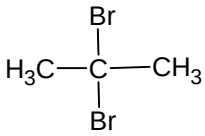
(B) 

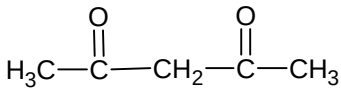
(C) 

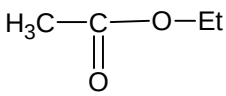
(D) 

(E) 

(F) 

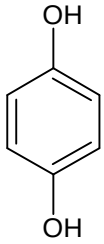
(G) 

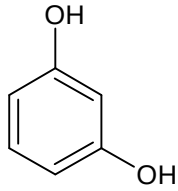
(H) 

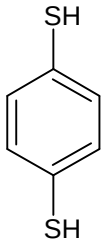
(I) 

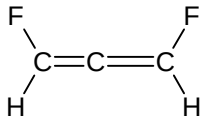
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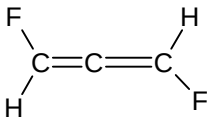
43. Find out no of compound which has non zero Dipole moment

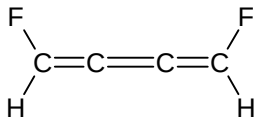
(A) 

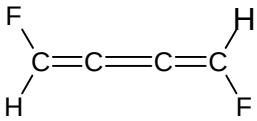
(B) 

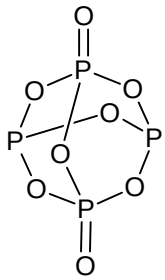
(C) 

(D) 

(E) 

(F) 

(G) 

(H) 

(I) PCl_2F_3

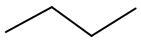
(J) I_2Cl_6

Space for rough work

44. Find out no of correct statements.
(A) $N_2 > N$ (Ist ionization Energy)
(B) $O_2 > O$ (Ist ionization Energy)
(C) $H_2 > H$ (Ist ionization Energy)
(D) $N_2^+ > N_2^-$ (Stability)
(E) $O_2 > O$ (Ist Electron affinity)
(F) $H_2 > H$ (Ist Electron affinity)
(G) $Na_2CO_3 > Li_2CO_3$ (Thermal stability)
(H) No of co-ordinate bond in P_4O_{10} is 4
(I) No of π bond in P_4O_8 is 2

45. Find out no of compounds which show mutarotation

(A) Cellobiose
(4 – O – (β – D – glucopyranosyl) – D – glucopyranose)
(B) Maltose
(4 – O – (α – D – glucopyranosyl) – D – glucopyranose)
(C) Lactose
(4 – O – (β – D - galactopyranosyl) – D – glucopyranose)
(D) Gentiobiose
(6 – O – (β – D - glucopyranosyl) – D - glucopyranose)
(E) Sucrose
(α – D – glucopyranosyl – β – D - fructofuranoside)
(F) Methyl α – D – galactopyranoside.
(G) α –D – allopyranose
(H) β – D – glucopyranose
(I) α – D - fructofuranose

46.  $\xrightarrow[h\nu]{Cl_2}$ Products

Find out no of correct statements about above reaction.
(A) No of monochlorinated products are three
(B) No of optically active monochlorinated products are two
(C) No of optically inactive monochlorinated product is one
(D) No of dichlorinated products are ten
(E) No of optically active dichlorinated product are six
(F) No of optically inactive dichlorinated products are 4
(G) No of meso dichlorinated products are two
(H) No of dichlorinated products are six
(I) No of optically active dichlorinated products are seven
(J) No of meso dichlorinated products is one

Space for rough work

Mathematics

PART – III

SECTION – A
Straight Objective Type

This section contains **08 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

47. If $S_n = 3 + \frac{1+3+3^2}{3!} + \frac{1+3+3^2+3^3}{4!}$..... upto n – terms

Then the value of $\left[\lim_{n\to\infty} S_n\right]$ is, (where [.] represent G.I.F)

(A) 6 (B) 7
(C) 8 (D) 9

48. Let f(x) be a polynomial of degree 4,
with $f(2) = -1, f'(2) = 0, f''(2) = 2, f'''(2) = -12, f''''(2) = 24$, then the value of $f''(1)$ is

(A) 24 (B) 26
(C) 28 (D) 30

49. A person whose hobby is tossing a fair coin is to score one point for every tail and two points for every head. The person goes on tossing the coin, till his score reaches 100 or exceeds 100. Then the probability that his score attains exactly 100 is,

(A) $\frac{2}{3} - \frac{1}{3.2^{100}}$ (B) $\frac{2}{3} + \frac{1}{3.2^{100}}$
(C) $\frac{2}{3} - \frac{1}{2^{100}}$ (D) $\frac{2}{3} + \frac{1}{2^{100}}$

50. Let $f(x) = x^5\left[\frac{1}{x^3}\right], x \neq 0$ & $f(0) = 0$ (where [.] represent G.I.F), Then $\lim_{x\to 0} f(x)$

(A) Does not exist (B) 1
(C) -1 (D) None of these

Space for rough work

51. In an examination of the maximum marks for each of the three papers are 50 each and maximum marks for the fourth paper is 100. Then the number of ways in which a candidate secure 60% marks in aggregate is
(A) 110551 (B) 40375
(C) 6176 (D) None of these

52. If the solution of the differential equation $e^{\frac{x}{y}(1-y^2)}\left[y\frac{dx}{dy}-x\right]+\left[xy^2+y^3\frac{dx}{dy}\right]=0$ is

$Ae^{xy}+Be^{\frac{x}{y}}+C=0$, Then A + B is
(A) -1 (B) 0
(C) 2 (D) None of these

53. If the line $x+y+1=0$ and $y-2x+5=0$ are tangents to a parabola whose focus at (1, 2), then the equation of normal to the parabola through $\left(\frac{29}{17},-\frac{14}{17}\right)$ is,
(A) $2x+4y-1=0$ (B) $y-3x+2=0$
(C) $4x+y-5=0$ (D) None of these

54. If $a+b+c=5, a^2+b^2+c^2=12$ and $a^3+b^3+c^3=25$ Then the value of $a^4+b^4+c^4$ is
(A) $\frac{251}{6}$ (B) $\frac{253}{6}$
(C) $\frac{255}{6}$ (D) $\frac{257}{6}$

Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

55. If C_1 and C_2 are two concentric circles of radii 4 and 5 respectively with centre at origin. A tangents is drawn to circle C_2 at a given point 'P'. This tangent is the diameter of a variable circle C_3, which touches C_1 externally. Also this tangent meet C_3 in Q & R. If the diameter QR subtends a constant non – zero angle at a fixed point T on line joining O & P. Then the locus of the point 'T' for various positions of P on C_2 is
(A) $x^2+y^2=2$ (B) $x^2+y^2=4$
(C) $x^2+y^2=64$ (D) $x^2+y^2=36$

Space for rough work

56. If g(x) is a continuous function such that $\int_0^x g(t)dt \to \infty$, as $|x| \to \infty$, Then the value of K for which line y = Kx intersect the curve $\int_0^x g(t)dt = 2 - y^2$

(A) -1 (B) $\sqrt{2}$
(C) 3 (D) 1

57. ABCD is a regular tetrahedron. P & Q are the mid – points of the edges AC and AB respectively, G is the centroid of the face BCD and θ is the angle between the vectors $\overrightarrow{PG}$ and $\overrightarrow{DQ}$, then

(A) The angle between $\overrightarrow{AB}$ and $\overrightarrow{CD}$ is 90^0

(B) The angle θ is $\pi - \cos^{-1}\left(\frac{5}{3\sqrt{3}}\right)$

(C) The angle θ is $\pi - \cos^{-1}\left(\frac{5}{6\sqrt{3}}\right)$

(D) The angle between $\overrightarrow{AB}$ and $\overrightarrow{CD}$ is 120^0

58. If $P = \sum_{r=3n}^{r=4n-1}\left[\frac{r^2+13n^2-7rn}{n^3}\right]$

& $Q = \sum_{r=3n+1}^{r=4n}\left[\frac{r^2+13n^2-7rn}{n^3}\right]$ Then

(A) $P > \frac{5}{6}$ (B) $Q < \frac{5}{6}$
(C) $Q > \frac{5}{6}$ (D) $P < \frac{5}{6}$

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 59 & 60

If n be a natural number define polynomial $f_n(x)$ of nth degree as follows

$f_n(\cos\theta) = \cos n\theta$

i.e $f_2(x) = 2x^2 - 1$ $\quad f_3(x) = 4x^3 - 3x,$

Then

59. $\left(x+\sqrt{x^2-1}\right)^{10} + \left(x-\sqrt{x^2-1}\right)^{10}$ is equal to

(A) $f_{10}(x)$ (B) $f_{11}(x) + f_9(x)$

(C) $f_{11}(x)$ (D) $2f_{10}(x)$

60. $f_6(x)$ is equal to

(A) $36x^6 - 48x^4 + 18x^2 - 5$ (B) $32x^6 - 48x^4 + 18x^2 - 1$

(C) $36x^6 - 45x^4 + 18x^2 - 8$ (D) $36x^6 - 48x^4 + 18x^2 - 7$

Paragraph for Questions 61 & 62

If the integrand is a rational function of x and fractional powers of a linear fractional function of the form $\frac{Ax+B}{Cx+D}$, then rationalization of the integral is affected by the substitution

$\frac{Ax+B}{Cx+D} = t^m$

61. If $\int \frac{(2x-3)^{\frac{1}{2}}}{(2x-3)^{\frac{1}{3}}+1}dx = 3\left[\frac{1}{7}(2x-3)^{\frac{7}{6}} - \frac{1}{5}(2x-3)^{\frac{5}{6}} + \frac{1}{3}(2x-3)^{\frac{1}{2}} - (2x-3)^{\frac{1}{6}} + g(x)\right] + C$

Then g(x) is,

(A) $\tan^{-1}(2x-3)^{\frac{1}{6}}$ (B) $(2x-3)^{\frac{1}{6}}$

(C) $3\tan^{-1}(2x-3)^{\frac{1}{6}}$ (D) $4(2x-3)^{\frac{1}{6}}$

Space for rough work

62. If $\int \frac{dx}{\sqrt[3]{(x+1)^2(x-1)^4}} = K\sqrt[3]{\frac{1+x}{1-x}} + C$,

Then K is equal to,

(A) $\frac{2}{3}$ (B) $-\frac{3}{2}$

(C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Paragraph for Questions 63 & 64

Consider a plane $P \equiv 2x + y - z - 5 = 0$, a line $\frac{x-3}{2} = \frac{y+1}{-3} = \frac{z-2}{-1}$ and a point $A(3,-4,1)$. L_2 is a line passing through A intersecting L_1 an parallel to the plane P.

63. Plane containing L_1 and L_2 is,

(A) Parallel to yz - plane (B) Parallel to x-axis

(C) Parallel to y-axis (D) Passing through origin

64. Line L_1 intersect plane P at Q and xy – plane at R, then the volume of tetrahedron OAQR is, (where 'o' is origin)

(A) 0 (B) $\frac{14}{3}$

(C) $\frac{3}{7}$ (D) $\frac{7}{3}$

SECTION – C
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

65. If the set of 'K' for which two distinct chords of the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$ passing through (2, -1) are bisected by the line x + y = K is [a, b], then (a + b) is……..

Space for rough work

66. If $ax^3 - cx + b \geq 0 \ \forall x \in R^+ - \{0\}$, where a, b, c $\in$ R^+. Then the minimum value of $\left(\frac{27ab^2}{c^3}\right)$ is....

67. If x, y, z are positive real number, such that x + y + z = 1, If the minimum value of $\left(1+\frac{1}{x}\right)\left(1+\frac{1}{y}\right)\left(1+\frac{1}{z}\right)$ is K^2, then $|K|$ is.....

68. Let A be a square matrix of order 3 satisfies $A^3 - 6A^2 + 12A - 8I = 0$ & $B = A - 2I$
If $|A| = 8$ & $\left|\text{adj}\left(I - 2A^{-1}\right)\right| = K$, then $[K]$ .....where [.] represents G.I.F

69. If $\frac{9a}{\cos\theta} + \frac{5b}{\sin\theta} = 56$ and $\frac{9a\sin\theta}{\cos^2\theta} - \frac{5b\cos\theta}{\sin^2\theta} = 0$
If the value of $\left[(9a)^{\frac{2}{3}} + (5b)^{\frac{2}{3}}\right]^3 = (8K)^2$ Then K is.....

Space for rough work